STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 14: Waves

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Section A - 1-mark questions - 8 x 1 = 8 marks

1. Transverse oscillation relative to wave travel? [1]
2. Sound in air type? [1]
3. Wave speed relation? [1]
4. Wave number k? [1]
5. Superposition principle? [1]
6. Reflection at fixed end? [1]
7. Beat frequency? [1]
8. What is amplitude? [1]

Section B - 2-mark questions - 6 x 2 = 12 marks

9. f=50Hz, lambda=2m: speed? [2]
10. v=340m/s, f=170Hz: wavelength? [2]
11. 256Hz and 260Hz beats? [2]
12. Wavelength halves at fixed speed: frequency? [2]
13. Reflection at free end: inversion? [2]
14. Wave y=A sin(kx-omegat) travels which way? [2]

Section C - 3-mark questions - 4 x 3 = 12 marks

15. v=24m/s, lambda=.8m: f and period? [3]
16. Displacements +3 cm and -1 cm overlap: resultant? [3]
17. 300Hz and 304Hz: beat frequency and period? [3]
18. How do waves transfer energy without permanent medium transport? [3]

Section D - 4-mark questions - 2 x 4 = 8 marks

19. Wave travels 12m in .4s with f=5Hz: speed/wavelength? [4]
20. Explain beats for 440Hz and 442Hz. [4]

End of paper - check working, units, signs and significant figures.